COVID-19 Global Impact Dashboard

Data Analytics Project Report

Tools: Python • Pandas • Matplotlib • Microsoft Excel • Power BI / Tableau (dashboard build step)

Dataset: Our World in Data COVID-19 public dataset

Prepared dataset coverage: 2020-01-03 to 2023-10-23

Project purpose

Explore reported COVID-19 cases, deaths, vaccination indicators, and changes over time. The project emphasizes data quality, careful comparisons, and communicating limitations clearly.

1. Dataset and Scope

The original uploaded source contains 350,085 rows, 255 locations, and 16 selected fields in this analysis extract. To keep the repository upload manageable, the included CSV is filtered to the World aggregate, continent aggregates where available, and 12 selected countries: United States, India, Brazil, United Kingdom, France, Germany, Italy, Japan, Canada, Australia, South Africa, Indonesia. It is not a replacement for the full source dataset.

2. Data preparation

The workflow parses dates, converts analytical measures to numeric values, sorts records by location and date, and preserves missing values rather than automatically treating them as zero. COVID-19 reporting is incomplete and can be revised; missing values therefore remain visible for interpretation.

3. Visual analysis

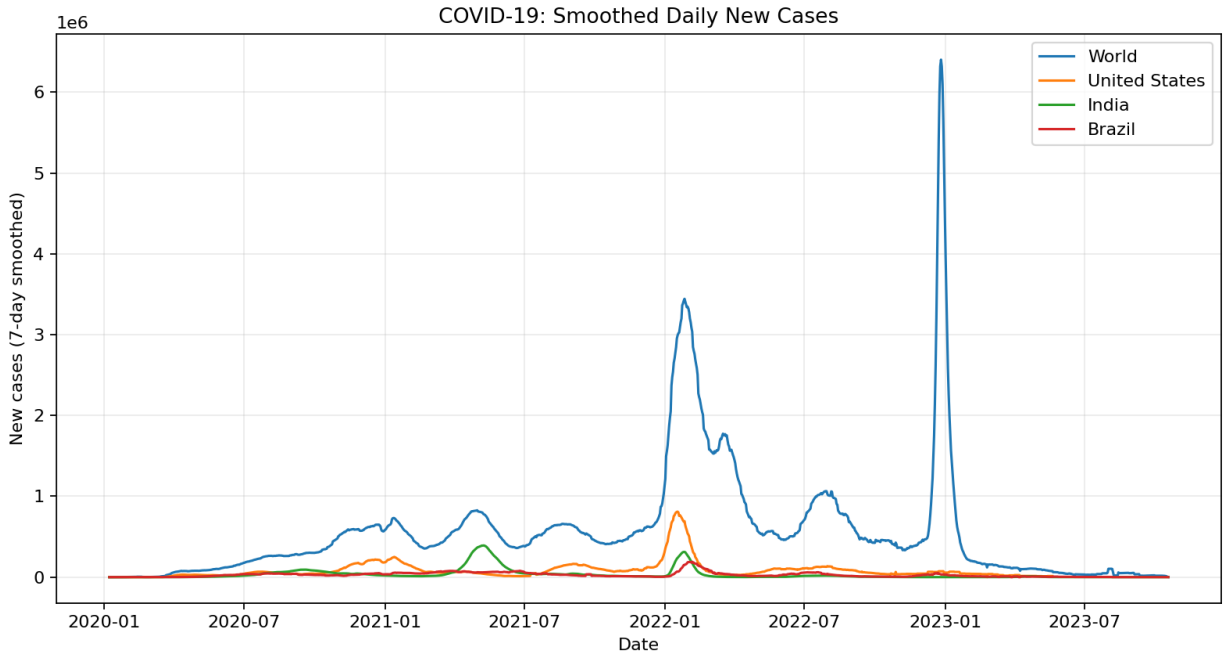


Figure 1. Smoothed reported new cases for selected locations. Use this view to discuss broad waves and reporting changes, not as a direct measure of all infections.

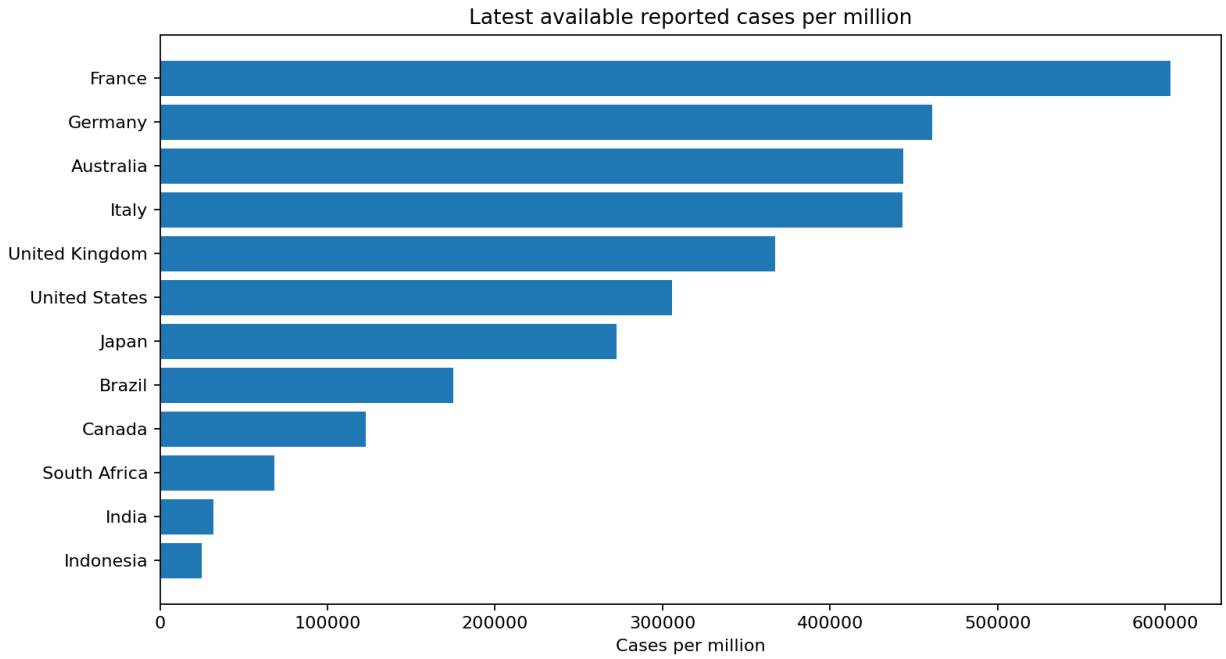


Figure 2. Latest available reported cases per million in the prepared country extract. The latest non-missing date can vary across locations, so verify dates before making a strict ranking.

4. Key observations

• The time series has multiple periods of increased reported cases. • Per-million measures are more suitable than raw totals for population-size comparisons, though testing and reporting differences still matter. • Vaccination measures have uneven coverage and may have different last-reported dates. • Reported counts are not the same as all infections or deaths.

5. Dashboard design plan

Recommended Power BI/Tableau page: KPI cards for latest available global cumulative cases and deaths; line charts for smoothed new cases and deaths; a country comparison bar chart using per-million values; and date, continent, and location filters. Avoid summing cumulative totals across dates.

6. Limitations and reproducibility

This report describes a historical public dataset extract, not live surveillance. Some values are missing or revised, and country reporting practices differ. The repository includes the filtered CSV for convenience. For full coverage, download the current/historical source file from Our World in Data and rerun the script with the downloaded CSV after adjusting the input path/column scope.

Source: Our World in Data, COVID-19 dataset documentation:
https://github.com/owid/covid-19-data/tree/master/public/data